

PRASHANT GAUTAM

Civil Engineer | Transportation Infrastructure & Resilience

077bce104.prashant@pcampus.edu.np • +977-9865442326 • www.prashant-gautam.com.np

PROFESSIONAL SUMMARY

Licensed civil engineer with hands-on expertise in mountain transportation infrastructure, resilient design, and post-fire structural diagnostics. Field-experienced in site supervision, surveying, and project coordination. Seeking graduate research opportunities to advance resilience-focused transportation systems in challenging environments.

EDUCATION

Bachelor's in Civil Engineering

2021 - 2025

Pulchowk Campus, Institute of Engineering (IOE), Tribhuvan University (TU)

- Regular program scholarship recipient.

+2 Science

2018 - 2020

Ambition Academy Secondary School, National Examination Board (NEB)

- Recipient of scholarship provided by Kathmandu Metropolitan City.

RESEARCH INTERESTS

- **Transportation Infrastructure Resilience:** Post-fire assessment, structural health monitoring, and resilient design of tunnels, mountain highways, and critical transit corridors in hazard-prone regions.
- **Mountain Transportation Systems:** Drainage design, slope stability, and infrastructure optimization for steep terrain. Natural-hazard resilience and transportation asset performance in mountainous environments.
- **Infrastructure Diagnostics:** Non-destructive testing (NDT), material forensics, and condition assessment to support repair vs. demolition decisions for aging infrastructure.

RELEVANT EXPERIENCE

Civil Engineer (Site Supervision & Measurement)

Aug 2025 – Present

Dibyashwori Nirman Sewa and various JVs linked.

- Supervise construction of ~12,000 stairs and mountain trail work (alignment, drainage, masonry soling) in steep terrain.
- Manage site layouts, surveying, and alignment verification using Total Station and theodolite.
- Perform precise quantity take-offs and on-site measurements for billing and engineering transparency.
- Ensure compliance with structural design standards and approved engineering drawings.

Civil Engineer (Structural Assessment)

Oct 2025 – May 2026

Institute of Engineering Consultancy Service (IOECS)

- Led post-fire detailed damage assessments (DDA) on 44 federal and central government buildings across Kathmandu and Pokhara to evaluate fire-induced structural compromise and safety risk.
- Executed non-destructive and destructive testing: concrete core sampling, carbonation depth analysis, UPV, rebound hammer to determine residual material strength and durability.
- Assessed thermal-induced degradation using ETABS structural modeling to support repair vs. demolition recommendations.
- Developed comprehensive cost estimates and Bills of Quantities (BOQ) for rehabilitation works, adhering to Government of Nepal norms.

Civil Engineer Intern

Sep 2024 – June 2025

D.S. Nirman Sewa

- Managed independent site layouts, surveying, and material tracking on active infrastructure projects.
- Performed quantity take-offs to ensure materials remained within budget and schedule.

CORE COMPETENCIES

- **Engineering:** Seismic design (ETABS), structural diagnostics, post-fire assessment, damage classification, repair planning
- **Field Investigation:** NDT (UPV, Rebound Hammer, Core Sampling), surveying (Total Station, Theodolite, Auto Level)
- **Software:** ETABS, AutoCAD, SW-DTM, Excel (advanced)
- **Design Standards:** IS 1893:2016 (Seismic), IS 13920 (Ductile Detailing), Nepal National Building Code (NBC)

ACADEMIC CAPSTONE PROJECT

Seismic Resistant Design of Multi-Storied RCC Commercial Building

2024–2025

- Performed comprehensive structural modeling and dynamic analysis using ETABS for B+G+9 storied building.
- Designed structural members per IS 1893:2016 and IS 13920 (ductile detailing) for earthquake resilience.
- Conducted equivalent static and response spectrum analysis per Nepal National Building Code.

PROFESSIONAL DEVELOPMENT & LEADERSHIP

- **Technical Certification:** ADB-sponsored training on Road Slope Stability and Bioengineering (RCIP-AF TA-10113, Nepal).
- **Team Leadership:** Project group leader; Academic Class Representative.
- **Magazine Editor:** Coordinator and editor, 'The Bridge' Magazine (CESS-Nepal).
- **Exhibition Organizer:** Primary coordinator, campus health camp; core technical coordinator, 'AADHAR-2081' Civil Engineering Exhibition.